

10.3 The challenges of food supply

This chapter will explain:

- ★ the physical (natural) and human factors that negatively affect food supply
- ★ strategies to increase food supply
- ★ the problems caused by food insecurity
- ★ the sustainable strategies and techniques used to manage desertification and soil erosion
- ★ detailed specific example: food insecurity in Nigeria.

The physical (natural) and human factors negatively affecting food supply

A wide range of factors combine to influence agricultural land use and practices on farms. These can be placed under the general headings of physical (or natural) factors and human factors, including economic, social/cultural and political factors.

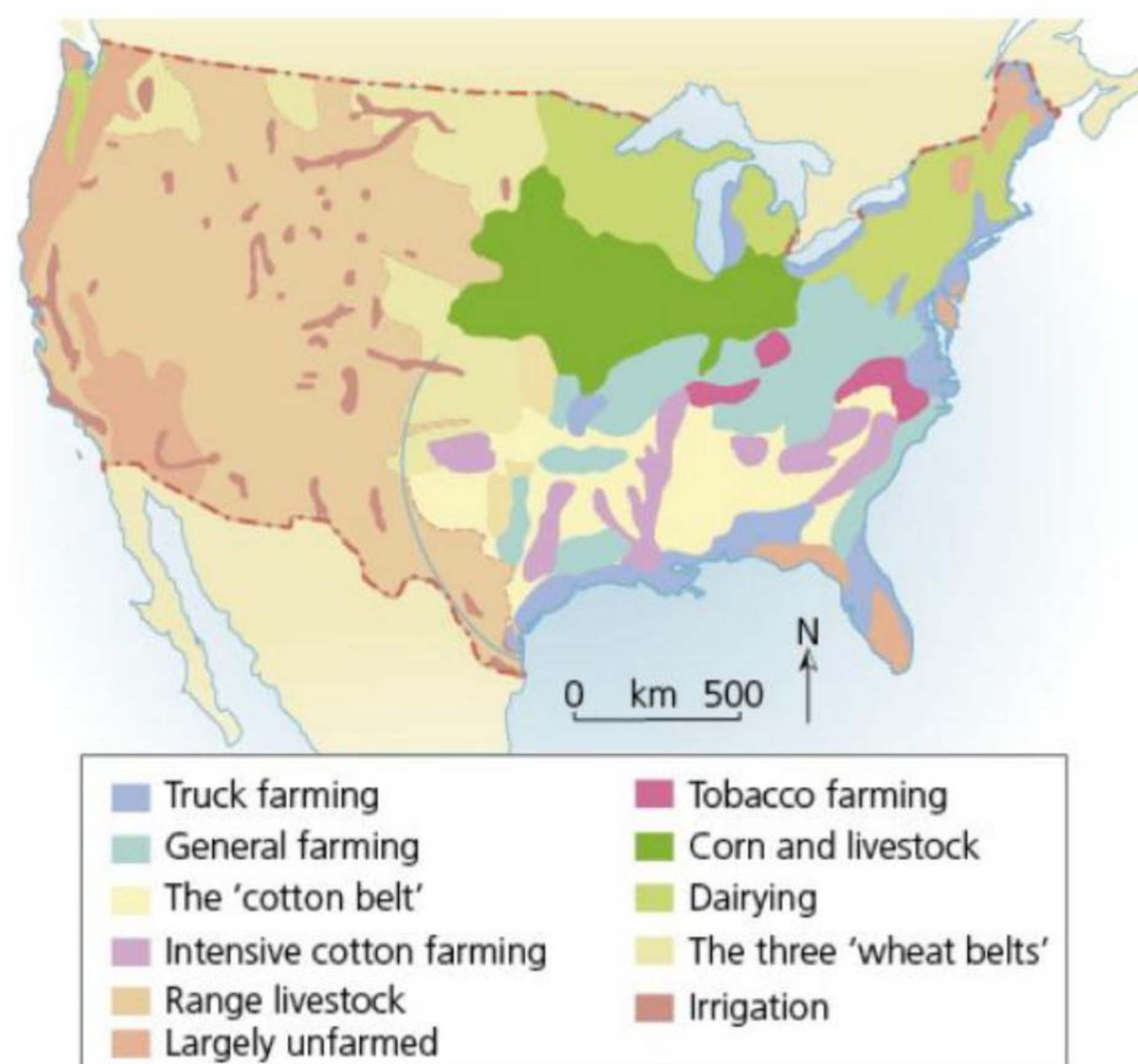
Physical factors

Physical factors set broad limits on:

- » whether agriculture is feasible in an environment
- » if agriculture is possible, how productive it might be.

In addition to these factors, the general agricultural productivity of an environment might be drastically altered for a certain time period by extreme weather or tectonic events.

Northern America, for example, has many different physical environments. This allows a wide variety of crops to be grown and livestock kept. New technology and high levels of investment have steadily extended farming into more difficult environments. Irrigation has enabled farming to flourish in the dry southwest, while new varieties of wheat have pushed production northwards in Canada. However, the physical environment remains a big influence on farming. There are certain things that technology and investment can do little to alter. So relief, climate and soils set broad limits as to what can be produced. This leaves the farmer with some choices, even in difficult environments. The farmer's decisions are then influenced by human (economic, social/cultural and political) factors.



▲ **Figure 10.17** The agricultural regions of the USA

[Figure 10.17](#) shows the **agricultural regions** of the USA. Look at relief and climate maps of the USA online or in an atlas to see how agricultural regions vary according to different physical conditions.

Temperature is a critical factor in crop growth as each type of crop requires a minimum growing

temperature and a minimum growing season. Latitude, altitude and distance from the sea are the major influences on temperature. Precipitation is equally important. This is not just the annual total, but the way it is distributed throughout the year. Long, steady periods of rainwater to infiltrate into the soil are best, making water available for crop growth. In contrast, short heavy downpours can result in surface runoff, leaving less water available for crop growth and causing soil erosion.

Soil type and soil fertility have a huge impact on **agricultural productivity**. Areas that have never been cleared for farming are often ignored because soil fertility is poor or is perceived to be poor. In some regions wind can have a serious impact on farming, for example causing bush fires in some US states such as California. Locally, the land’s aspect (the direction it faces) and the angle of slope may also be important factors in deciding how to use the land.

Cotton, for example, needs a frost-free period of at least 200 days. Rainfall should be over 625 mm a year, with not more than 250 mm in the autumn harvest season. Cotton production is now highly mechanised. Irrigation has allowed cotton to flourish in the drier western states of California, New Mexico and Texas. In contrast, the cotton-growing area has fallen considerably in the southern states. A crop pest called the boll weevil, which caused great destruction to cotton crops in the past, has been a big factor in the diversification of agriculture in the southern states.

In contrast, corn is grown further north than cotton. Corn needs a growing season of at least 130 days. For the crop to ripen properly, summer temperatures of 21°C are needed, with warm nights. Precipitation should be over 500 mm, with at least 200 mm in the three summer months.

In Canada, farming is severely restricted by climate. Less than 8 per cent of the total area of the country is farmed. Seventy per cent of Canada lies north of the **thermal limit for crop growth**. Other high-latitude countries such as Russia, Sweden and Finland also face considerable climatic restrictions on agriculture.

Water is vital for agriculture. **Irrigation** is an important factor in farming in many parts of the world. [Table 10.2](#) compares the main types of irrigation. This is an example of the **ladder of agricultural technology**, with surface irrigation being the most traditional method and subsurface (drip) irrigation the most advanced technique. [Figure 10.18](#) shows an irrigation canal in northern Spain. Irrigation canals of this size can transport large volumes of water, which can then be filtered down to smaller volumes so that a large area of land can be irrigated.

▼ **Table 10.2** The types of irrigation

Type of irrigation	Efficiency (%)
Surface: Used in over 80 per cent of irrigated fields worldwide:	
a Furrow: A traditional method; cheap to install; labour-intensive; high water losses; susceptible to erosion and salinisation	20–60
b Basin: Cheap to install and run; needs a lot of water; susceptible to salinisation and waterlogging	50–75
Aerial (using sprinklers): Used in 10–15 per cent of irrigation worldwide; costly to install and run; low-pressure sprinklers preferable	60–80
Sub-surface (‘drip’): Used in 1 per cent of irrigation worldwide; high capital costs; sophisticated monitoring; very efficient	75–95

Human factors

The human factors influencing agriculture can be divided into economic, social/cultural and political.

Economic

Economic factors include transport, markets, capital and technology. The cost of growing different crops or keeping different livestock varies. The market prices for agricultural products will also vary and can change from year to year. The necessary investment in buildings and machinery can mean that some changes in farming activities are very expensive. These would be more difficult to achieve than other, cheaper changes. Thus it is not always easy for farmers to react quickly to changes in consumer demand.



▲ **Figure 10.18** An irrigation canal in northern Spain

In most countries there has been a trend towards fewer but larger farms. Large farms allow economies of scale to operate, which reduce the unit costs of production. As more large farms are created, small farms find it increasingly difficult to compete and make a profit. Selling to a larger neighbouring farm may be the only economic solution. The EU is an example of a region where average farm size varies significantly. Those countries with a large average farm size generally have more efficient agricultural sectors than countries with a small average farm size.

Agricultural technology is the application of techniques to control the growth and harvesting of animal and vegetable products. The development and application of agricultural technology requires investment and thus it is an economic factor. The status of a country's agricultural technology is vital for its food security and other aspects of its quality of life. The transfer of agricultural technology from more advanced to less advanced countries is therefore an important form of aid.

Social/cultural factors

What a particular farm and neighbouring farms have produced in the past can be a significant influence on current farming practices. There is a tendency for farmers to stay with what they know best and often a sense of transgenerational responsibility to maintain a family farming tradition. Tradition matters more in some farming regions than others.

Land tenure means the ways in which land is or can be owned. In the past inheritance laws have had a huge impact on the average size of farms. In some countries it has been the custom on the death of a farmer to divide the land equally between all his sons, but rarely between daughters. Also, dowry customs may include the giving of land with a daughter on marriage. The reduction in the size of farms by these processes often reduces them to operating at only a subsistence level.

In most societies women have very unequal access to, and control over, rural land and associated resources. It is now generally accepted that those societies with well-recognised property rights are also the ones that thrive best economically and socially.

Political factors

The influence of government on farming has steadily increased in many countries. For example, in the USA the main parts of government farm policy over the past half-century have been as follows:

- » Price support loans: Loans that help farmers until they sell their produce.
- » Production controls: These limit how much surplus a farmer can produce of crops.
- » Income supplements: These are cash payments to farmers for major crops in years when market prices fail to reach certain levels.

Thus the decisions made by individual farmers are heavily influenced by government policies such as those listed above. An agricultural policy can cover more than one country, as evidenced by the EU's Common Agricultural Policy.

Adverse influences on food supply and distribution

[Table 10.3](#) summarises some of the current adverse influences on food supply and distribution.

▼ **Table 10.3** Adverse influences on global food production and distribution

Nature of adverse	Effect of adverse influence
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influence	
Economic	● Demand for cereal grains has outstripped supply in recent years ● Rising energy prices and agricultural production and transport costs have pushed up costs all along the farm-to-market chain ● Serious underinvestment in agricultural production and technology in LICs has resulted in poor productivity and underdeveloped rural infrastructure ● The production of food for local markets (Figure 10.19) has declined in many LICs as more food has been produced for export
Ecological	● Significant periods of poor weather and a number of severe weather events have had a major impact on harvests in key food-exporting countries ● There have been increasing problems of soil degradation in both HICs and LICs ● Declining biodiversity may impact food production in the future
Sociopolitical	● The global agricultural production and trading system, built on import tariffs and subsidies, creates great distortions, favouring production in HICs and disadvantaging producers in LICs ● There is an inadequate international system of monitoring and deploying food relief ● There are disagreements over the use of transboundary resources such as river systems and aquifers ● Major international conflicts can significantly decrease the supply of important food products to the global market



▲ **Figure 10.19** Selling local produce in a Vietnamese food market

Climate loss in food production

Climate loss is pre-harvest food loss. These are the losses farmers experience from:

- » not planting a crop, or at least not the desired crop
- » changing crops due to weather
- » extreme weather reducing or wiping out a crop before harvest
- » increased predation, pests and diseases that are due to climate change.

The UN Food and Agriculture Organization report 'The Impact of Disasters on Agriculture and Food Security 2023' stated that over the previous 30 years, an average of US\$123 billion per year was lost due to disasters in agricultural crop and livestock production:

- » Losses were highest in LICs and lower MICs, totalling as much as 15 per cent of their agricultural GDP.
- » Small island states were more affected than average.

Activities

- 1 List the main physical factors that can influence farming.
- 2 Summarise the information presented in [Table 10.3](#).
- 3 Why has the size of farms steadily increased in many agricultural regions?
- 4 Briefly state the importance of advances in agricultural technology.
- 5 Give an example of how a social/cultural factor can have an impact on farming.
- 6 How can political factors influence farming?

The problems caused by food insecurity

The definition of **food security** is when all people at all times have access to sufficient, safe, nutritious food to maintain a healthy and active life. Food security has four components:

- » Availability: This is the existence of sufficient food within a community. It can become an issue when resources decline, for example water for irrigation and the degradation of soil.
- » Access: Access to food can be affected by factors such as rising prices, household proximity to food retailers and poor infrastructure.
- » Utilisation: Food needs to be of sufficient quality to provide the required levels of nutrition and households need the necessary knowledge and tools to properly select, prepare and store foods.
- » Stability: Access, availability and utilisation of food needs to remain relatively stable over time.

The total amount of food produced around the world today is enough to provide everyone with a healthy diet. The problem is that while some countries produce a food surplus or have enough money to buy it elsewhere, other countries are in food deficit and lack the financial resources to buy enough food abroad. About one in nine of the world's population remains chronically undernourished.

The current food crisis presents three fundamental threats. These threats are:

- » pushing more people into poverty
- » eroding development gains that have been achieved in many countries in recent decades
- » presenting a strategic threat by endangering political stability in some countries.

The impact of food insecurity has been felt most intensely in developing countries, where adequate food stocks to cover emergencies affecting food supply usually do not exist. However, developed countries have not been without their problems and are not immune to the physical problems that can cause food shortages. For example, in recent years both the USA and Australia have suffered severe drought conditions. However, developed countries invariably have the human resources to cope with such problems, so actual food shortages do not generally occur. However, in some HICs, an increasing number of people have had to rely on **food banks** to maintain a reasonable level of food intake.

Short-term and long-term effects

The effects of food shortages are both short-term and longer-term. **Malnutrition** can affect a considerable number of people, particularly children, within a relatively short period when food supplies are significantly reduced. With malnutrition, people are less resistant to disease and more likely to fall ill. Diseases connected to malnutrition include beriberi (vitamin B1 deficiency), rickets (vitamin D deficiency) and kwashiorkor (protein deficiency). People who are continually starved of nutrients never fulfil their physical or intellectual potential. Malnutrition reduces people's capacity to work, so land may not be properly tended or other forms of income successfully pursued. This is threatening to lock parts of the developing world into an endless cycle of ill-health, low productivity and underdevelopment.

The 2023 Global Hunger Index ([Figure 10.20](#)) noted:

- » little progress had been made in reducing hunger at a global scale since 2015
- » hunger remained serious or alarming in 43 countries
- » South Asia and Sub-Saharan Africa are the world regions with the highest hunger levels
- » the fight against hunger has been impeded by overlapping crises – Covid-19, multiple violent conflicts and climate disasters around the world.

Strategies and techniques used to increase food supply

The United Nations Environment Programme has argued that increasing food energy efficiency provides a critical path for significant growth in food supply without compromising environmental sustainability.

Options with short-term effects:

- 1 Price regulation on commodities and larger cereal stocks to decrease the risk of highly volatile prices.
- 2 To reduce/remove subsidies on biofuels to discourage farmers from using cropland for energy production rather than food production.

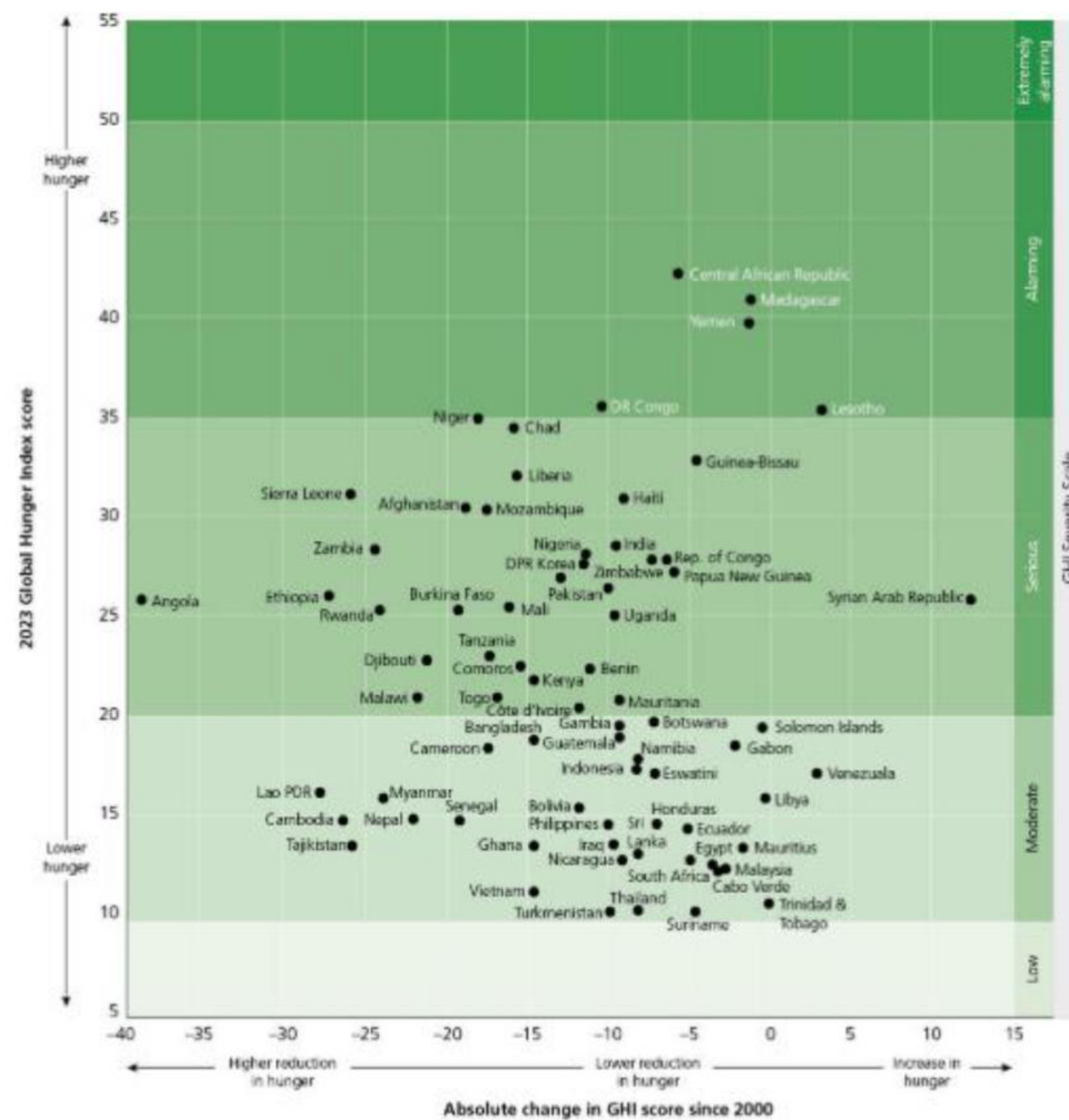
Options with mid-term effects:

- 1 To reduce the use of cereals and food fish in animal feed.
- 2 To support farmers in developing diversified eco-agricultural systems that provide critical ecosystem services (for example water supply and regulation) as well as adequate food to meet local and consumer needs.
- 3 To increase trade and improve market access by improving infrastructure and reducing trade barriers.

Options with long-term effects:

- 1 To limit global warming, including promoting climate-friendly agricultural production systems and land-use policies at a scale to help mitigate climate change.
- 2 To raise awareness of the pressures of increasing population growth and consumption patterns on sustainable ecosystem functioning.

See [page 254](#) for reference to the Green Revolution and the New Green Revolution.



▲ **Figure 10.20** The 2023 Global Hunger Index scores and progress since 2000

The role of food aid in improving food security

In the short term, and in some instances the medium term, food aid is absolutely vital to cope with food shortages ([Figure 10.21](#)). When disaster strikes, there is no alternative to this strategy. According to the NGO ActionAid there are three types of food aid:

- » Relief food aid, which is delivered directly to people in times of crisis.
- » Programme food aid, which is provided directly to the government of a country for sale in local markets (this usually comes with conditions from the donor country).
- » Project food aid, which is targeted at specific groups of people as part of longer-term development work.

The USA and the EU together provide about two-thirds of global food aid deliveries. At the international level the main organisations are the UN World Food Programme (WFP), the UN Food and Agriculture Organization (FAO) and the Food Aid Convention.



▲ **Figure 10.21** Food aid being delivered in Haiti

Food aid is vital for communities in many countries, particularly in Africa but also in parts of Asia and Latin America. However, it is not without controversy:

- » The NGO CARE has criticised the method of US food aid to Africa. CARE sees the selling of heavily subsidised US-produced food in African countries as undermining the ability of African farmers to produce for local markets, making countries even more dependent on aid to avoid famine. CARE wants the USA to send money to buy food locally instead.
- » Friends of the Earth says that a genetically modified rice, not allowed for human consumption and originating in the USA, has been found in food aid supplied to West Africa.

- » Food aid is very expensive, not least because of the high transport costs involved.

There have been recent concerns that food aid may be required for even more people in the future. In recent years, the term 'global food crisis' has been used more and more by the media. Steep increases in the price of food have caused big problems in a number of countries. Major protests about the price of food have taken place in countries including Haiti, Indonesia, the Philippines and Egypt. The World Bank has warned that progress on development could be destroyed by rapidly rising food costs.

Activities

- 1 List the four components of food security.
 - 2 What is malnutrition and how can it impact people and countries?
 - 3 Describe the different types of food aid.
 - 4 Produce a brief analysis of [Figure 10.20 \(page 263\)](#).
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Sustainable strategies and techniques used to manage desertification and soil erosion

The environmental and socioeconomic consequences of soil degradation are considerable. Such consequences can occur with little warning as damage to soil is often not perceived until it is far advanced.

Soil erosion

The increasing world population and the rapidly changing diet of hundreds of millions of people as they become more affluent is placing more and more pressure on land resources. Some soil and agricultural experts say that a decline in long-term soil productivity is already seriously limiting food production in the developing world.

A range of management strategies can be employed to reduce soil degradation. Many of these strategies can be subdivided into:

- » mechanical methods
- » cropping techniques.

Mechanical methods focus on preventing or slowing down the movement of rainwater downslope and reducing the impact of wind on the soil. Afforestation (planting trees on areas that have not previously been forested) or reforestation have been widely used over sizeable areas or as part of a wider package of measures. The selection of the areas to be forested is of crucial importance. Steeper slopes are the natural starting point but other factors must be taken into account, such as the predominant wind direction.

Contour ploughing is a tried and trusted technique that prevents or diminishes the downslope movement of water and soil. Such ploughing ensures that ridges and furrows are at right angles to the slope. Where slopes are too steep for contour ploughing, **terracing** ([Figure 10.22](#)) may be practised. Here the steep slope is converted into a series of flat steps with raised outer edges (bunds). The monsoon regions of Southeast Asia exhibit widespread terracing.



▲ **Figure 10.22** Terracing in a valley in the Andes Mountains, Peru

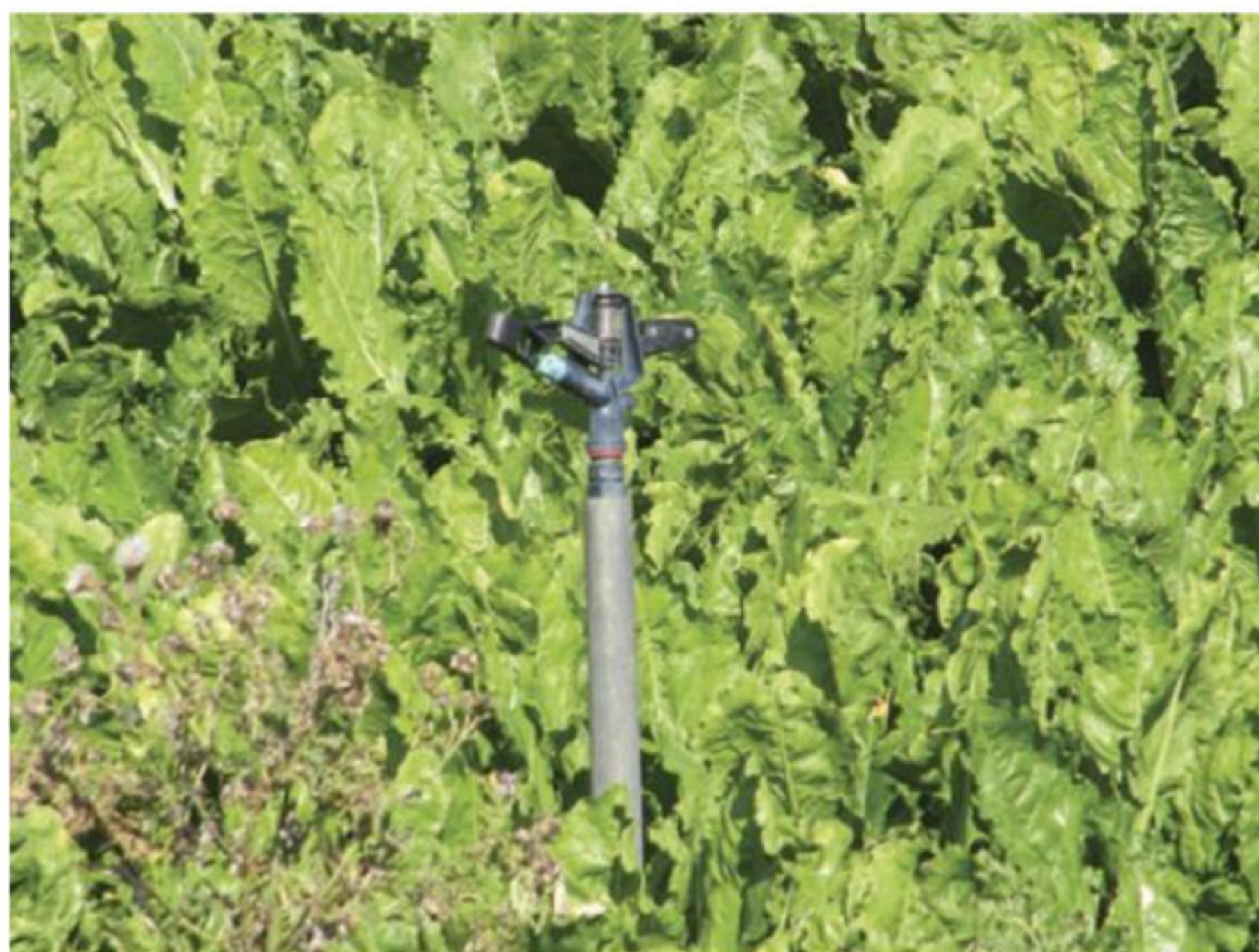
The planting of trees in **shelterbelts** and the use of hedgerows can do much to dissipate the impact of strong winds, reducing the wind's ability to disturb topsoil and erode particles.

Various cropping techniques can be employed to reduce soil degradation:

- » Converting land from arable to pastoral uses. The planting of grass helps to bind soil particles together, reducing the action of wind and rain.
- » Including grasses in crop rotations.
- » Leaving unploughed grass strips between ploughed fields.
- » Keeping a crop cover on the soil for as long as possible, thus minimising the 'bare soil' period.
- » Increasing the organic content of the soil by applying animal manure, compost or sewage sludge. This

enables soil to hold more water, preventing aerial erosion and stabilising soil structure.

- » Selecting and using farm machinery carefully, in particular avoiding where possible the use of heavy machinery on wet soils to prevent damage to the soil structure and using low ground pressure set-ups on machinery when possible.
- » Leaving the stubble and root structure in place after harvesting.
- » Using reduced or shallow cultivation to maintain or increase near-surface organic matter.
- » Shepherding livestock and moving forage areas to avoid overgrazing.
- » Using the most modern and efficient means of irrigation ([Figure 10.23](#)).



▲ **Figure 10.23** A sprinkler system irrigating crops in northern Spain

Many degraded environments require substantial investment to bring in realistic solutions. Such finance is beyond the means of many LICs. However, there may be a choice between low-cost and high-cost schemes, as [Table 10.4](#) illustrates.

▼ **Table 10.4** Technical and institutional costs in resource management by and for those in poverty

	High institutional costs	Low institutional costs
Relatively high technical costs	● Large-scale irrigation ● Arid or semi-arid land reforestation or pasture improvement ● Sodic or saline land reclamation ● Mangrove reforestation ● Integrated river-basin management ● Shaping or changing many transboundary resources, e.g. international rivers, air quality	● Small-scale hill irrigation ● Food crop systems on difficult soils ● Localised water-harvesting structures ● Centralised provision of energy services ● Solar energy for individual households ● Pipe sewer systems
	● Resettlement schemes ● Water-pollution reduction programmes ● Rural road maintenance ● Ocean fisheries management	● Emissions-reduction devices ● Improved public transport
Relatively low technical costs	● Aquifer management ● Protection of critical areas ● Coastal fisheries management ● Coral-reef management ● Pasture management ● Land-reform programmes ● Integrated pest management ● Wild game management	● Treadle pump irrigation ● Humid tropics reforestation ● Small water-harvesting systems ● Joint forest management regimes ● Improved cooking stoves and cooking energy for low-income families ● Sloping agricultural land technology (SALT) ● Small-scale quarrying ● Household-based sanitation systems

Managing salt-affected soils

Periodic soil testing and treatment, combined with proper management procedures, can improve the

conditions in salt-affected soils ([Figure 10.24](#)) that contribute to poor plant growth. There are three ways to manage saline soils:

- » Salts can be moved below the **root zone** by applying more water than the plant needs. This is known as the leaching requirement method.
- » Where soil moisture conditions dictate, the leaching requirement method can be combined with artificial drainage.
- » Salts can be shifted away from the root zone to locations in the soil, other than below the root zone, where they are not harmful. This is called managed accumulation. Sometimes, salt-tolerant crops may need to be selected in addition to managing soils.



▲ **Figure 10.24** Infertile saline soil in the south of France

The rise of no-till farming

The traditional practice of turning the soil before planting a new crop is a leading cause of soil degradation. An alternative is no-till farming, which minimises soil disruption. Here, farmers leave crop residue on the fields after harvest, where it acts as a mulch to protect the soil and provide nutrients. To sow seeds, the farmers use seeders that penetrate through the residue to the undisturbed soil below. This important sustainable approach to farming is spreading, but so far it has been mainly confined to major farming nations because of the high equipment costs involved in changing from traditional practices.

The advantages of no-till farming are that it reduces soil erosion, conserves water, improves soil health, cuts fuel and labour costs, reduces sediment and fertiliser pollution in lakes and streams, and sequesters carbon. The drawbacks, apart from costly new machinery, are greater reliance on herbicides, and that more nitrogen fertiliser may initially be required. There may be greater prevalence of weeds and other pests, and yield may be reduced.

Desertification

Desertification is the gradual transformation of habitable land into desert. It is arguably the most serious environmental consequence of soil degradation. Desertification is usually caused by climate change and/or by destructive use of the land. The natural causes of desertification include temporary drought periods of high magnitude and long-term climate change towards aridity. The main human causes are:

- » overgrazing
- » overcultivation
- » deforestation.

Desertification occurs when already fragile land in arid and semi-arid areas is over-exploited. It is a considerable problem in many parts of the world, for example on the margins of the Sahara Desert in Northern Africa and the Kalahari Desert in Southern Africa. In semi-arid areas such as the edge of the Kalahari Desert, a combination of low and variable precipitation, nutrient-deficient soils and heavy dependence on subsistence farming make soil degradation a significant threat. At present, 25 per cent of the global land territory and nearly 16 per cent of the world’s population are threatened by desertification. There are many ways of combating this, depending on the perceived causes ([Table 10.5](#)).

▼ **Table 10.5** The strategies for preventing desertification and their disadvantages

Cause of desertification	Strategies for prevention	Problems and drawbacks
Overgrazing	● Improved stock quality: Vaccination programmes and the introduction of better breeds enable yields of meat, wool and milk to be increased without increasing the herd size ● Better management: Reducing herd sizes and grazing over wider areas both reduce soil damage	● Vaccination programmes improve survival rates, leading to bigger herds ● Population pressure often prevents these measures

Overcultivation	● The use of fertilisers: These can double yields of grain crops, reducing the need to open up new land for farming ● New or improved crops: Many new crops or new varieties of traditional crops with high-yielding and drought-resistant qualities could be introduced ● Improved farming methods: The use of crop rotation, irrigation and grain storage can all increase, which would reduce pressure on land	● The cost to farmers can be high ● Artificial fertilisers may damage the soil ● Some crops need expensive fertiliser ● There is a risk of crop failure ● Some methods require expensive technology and special skills
Deforestation	● Agroforestry: This combines agriculture with forestry, allowing the farmer to continue cropping while using trees for fodder, fuel and building timber. Trees protect, shade and fertilise the soil ● Social forestry: Village-based tree-planting schemes involve all members of a community ● Alternative fuels: Oil, gas and kerosene can be substituted for wood as sources of fuel	● There is a long growth time before the benefits of trees are realised ● Expensive irrigation and maintenance may be needed ● Expensive, special equipment may be needed

Activities

- 1 What is soil erosion and to what extent is it a major global problem?
- 2 Describe three techniques commonly used to combat soil erosion.
- 3 Define desertification.
- 4 With reference to [Table 10.5](#), explain one strategy for preventing desertification for each of the three causes of desertification.

Detailed specific example

Food insecurity in Nigeria

With a population over 220 million, Nigeria is the most populous country in Africa. The country achieved lower-middle-income status in 2014. Agriculture is an important economic sector in Nigeria, contributing about 23 per cent to the country's GDP, and accounting for just over 50 per cent of all employment. However, about 37 per cent of the population live below the poverty line. In 2023, the Nigerian government declared a state of emergency on food security. In early 2024, the World Food Programme projected that 26.5 million people across Nigeria would face acute hunger in the June–August lean season.

Factors affecting food supply and food insecurity

Nigeria's food supply problems are due to a combination of physical and human factors. The country is affected by periodic floods and droughts that have a major impact on agricultural output. Such situations increase the vulnerability of populations, particularly in rural areas. Widespread flooding in the 2022 rainy season damaged more than 675,000 hectares of farmland, impacting heavily on national food supply. More extreme weather patterns influencing food security are expected in the future.

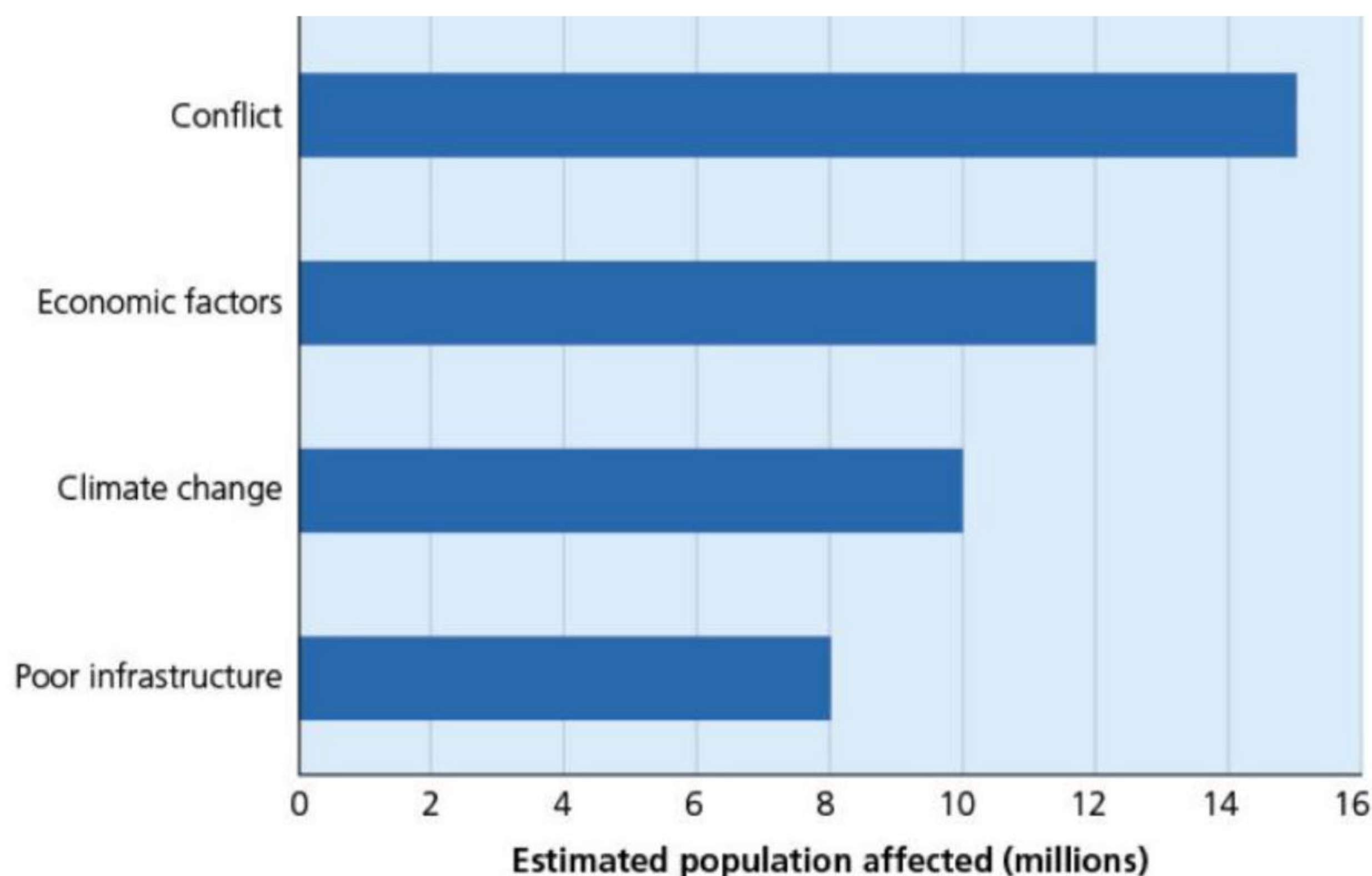
Land degradation and desertification have reduced production in a number of regions. Fifteen out of the 36 states of Nigeria are affected by desertification. Other factors impacting Nigeria's food crisis include the following:

- A relatively low level of irrigation, which limits agricultural production.
- Poor access to markets.
- The longstanding conflict in Nigeria's northeast region has displaced over 2 million people and created a particularly high level of food insecurity in this region. Agricultural production is frequently interrupted by attacks from insurgents, with important infrastructure often destroyed. More recent conflict has also affected the northwest and north-central regions.
- Clashes between herders and farmers over the use of land deter investment in the areas most affected.
- A high level of inflation is driving rising food prices. Food consumption makes up about 50 per cent of average expenditure for Nigerian households.

[Figure 10.25](#) illustrates a recent estimate of the main causes of the food crisis in Nigeria.

The Nigerian government sees the country's current food systems as being unsatisfactory and in urgent need of improvement. Agricultural yields in the country are some of the lowest in the world.

- Post-harvest losses are high, with more than 10 per cent of production lost post-harvest for several crops.
- Food storage facilities are often lacking in capacity and quality, resulting in a high level of food waste.
- Food processing centres often lack automation.
- Poor road infrastructure and connectivity causes delays in the transportation and distribution of food, resulting in poor food safety.
- Agricultural practices have contributed to a high level of soil degradation and erosion.



▲ **Figure 10.25** The primary causes of the food crisis in Nigeria

Problems caused by food insecurity

People who experience food insecurity are more likely to suffer malnutrition, which compromises the immune system and can result in chronic conditions such as diabetes and heart disease. Malnutrition stunts growth and impairs cognitive development. The long-term consequences can extend into adult life, continuing a cycle of poverty and ill-health. Malnutrition is at a high level in Nigeria, especially in the more remote conflict-affected areas in the northeast. In a number of regions, nutrition stabilisation centres have been essential to keeping children alive. Having a significant number of people in this situation places intense pressure on medical services and also affects labour productivity in agriculture and other economic sectors. Children are the most vulnerable to food insecurity.

Strategies to increase food supply

Food supply challenges exist in all parts of the country, but they vary in intensity across states and within states. Therefore there is a need for location-specific interventions. Recent strategies have built upon a variety of existing programmes and include:

- modernising agricultural infrastructure, including storage facilities and processing plants
- increasing investment in research and development
- expanding mechanisation and modern farming techniques
- improving access to credit and finance for farmers, including microfinance schemes
- strengthening agricultural education and training
- promoting sustainable agricultural practices
- diversifying agricultural crops to produce a wider range of food products
- investment in agro-processing, as the value added will move up the value chain. This will increase the value of Nigeria's agricultural exports.

Sustainable strategies need to result in higher incomes and reduced poverty among smallholder farmers. Scaling up more environmentally friendly practices, such as increasing the number of cropping cycles in a year and raising yields, will be essential to increasing food production in Nigeria. A number of national and international organisations have stressed the need for land tenure reforms in Nigeria. Farmers frequently face problems in the acquisition of suitable land for farming activities. In addition, increasing urbanisation and other built development has resulted in the sale of agricultural land for construction.

Nigeria has established a number of important agricultural policies over the years, but many have not delivered the desired objectives due to a combination of inadequate funding, poor coordination and arguably a lack of political will.

The World Food Programme

The World Food Programme (WFP) is playing an important role in combating malnutrition. It places a strong emphasis on providing specialised nutritious food to children under five at risk of malnutrition. In collaboration with the Nigerian government and other agencies, it has set up livelihoods programmes and income-generating activities such as helping people acquire skills and employment in areas such as food processing, aquaculture and vegetable gardening. WFP also provides:

- technical assistance to the national home-grown school feeding programme
- air transport to the humanitarian community working in hard-to-reach isolated areas
- internet connectivity to key agricultural advisors via its Emergency Telecommunications Service.

Activity

Discuss the factors affecting food supply and food insecurity in Nigeria.

